=== ALICE KNOWLEDGE BASE TEST DOCUMENT ===
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PROJECT HELIOS — Phase 2 Summary
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Project Codename: Helios
Lead Researcher: Dr. Elena Vasquez
Department: Advanced Propulsion Systems
Start Date: March 15, 2026
Target Completion: December 2026

The Helios project aims to develop a new plasma-based ion thruster for deep space missions. Phase 1 demonstrated a thrust efficiency of 73.4% using argon fuel at 2.1 kW input power. Phase 2 has been focusing on switching to xenon propellant which should theoretically achieve 82% efficiency at the same power level.

Key Milestones Achieved in Q2 2026:
- April 12: Xenon injection system prototype built
- May 3: First successful plasma ignition with xenon (duration: 47 seconds)
- May 28: Sustained 12-minute burn achieved at 1.8 kW
- June 19: Peak efficiency measurement: 78.9% at 2.0 kW (xenon)

Current Challenges:
1. Cathode erosion rate is 0.4 micrometers per hour — 3x higher than expected
2. Thermal management at the nozzle throat exceeds 1850 Kelvin during extended burns
3. Xenon flow regulation shows ±2.3% instability in microgravity simulation

Budget Status:
- Allocated: $4.2 million
- Spent (Q1-Q2): $1.85 million
- Remaining: $2.35 million
- Burn rate: $308,000 per month

Team Composition:
- 4 senior propulsion engineers
- 2 plasma physicists
- 3 mechanical engineers (thermal systems)
- 1 project coordinator (Maria Kowalski)
- 2 graduate interns (rotating, currently: James Chen and Sarah O'Brien)

Upcoming Decisions (Q3 2026):
- Whether to proceed with the magnetic nozzle redesign (estimated cost: $420,000)
- Selection of cathode material: lanthanum hexaboride vs barium-impregnated tungsten
- Decision on collaboration offer from ESA's Advanced Concepts Team

Personal Notes from Dr. Vasquez (July 10, 2026):
"The xenon results are promising but the cathode issue worries me. If we can't solve the erosion problem by September, we may need to request a 3-month extension and additional $600,000 in funding. The ESA collaboration could give us access to their vacuum chamber which handles 10^-9 torr — far better than our current 10^-7 torr capability."

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